

Direction change of redirected walking via a single shoe height change

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ABSTRACT

The authors explore the effect of proprioceptive system changes caused by foot height differences on reorientation walking by asking users to walk in a virtual environment with shoes of highly inconsistent heights. The results show that the height difference of the shoes has a great influence on the participants' walking direction in the virtual environment, under normal circumstances, the participant will lean to one side when walking, but when the height difference between the two shoes increased, participants were more likely to lean in the opposite direction.

Keywords: Redirected walking, the height of shoe; change direction

Index Terms: Human-centered computing~User studies • Human-centered computing~Virtual reality • Human-centered computing~Haptic devices

1 INTRODUCTION

Humans use visual, somatosensory, vestibular, and auditory cues to complete locomotor tasks [1]. However, Human perception is not always consistent with how humans act [2]. The redirected walking was inspired by the fact that a person wearing an eye mask was instructed to walk in a straight line but walked in an arc without knowing it [3]. At present, most of RDW techniques only used visual stimuli [4] and optionally spatial audio [5]. Some studies have explored redirected walking through Visuo-haptic, such as "Unlimited Corridor" and "Handrail", both of them can make users walk around virtual environments (VE) unlimitedly with touching walls although they are walking around circular walls within a limited space [6]. Dominik Schmidt et al. proposed the "Level-Ups", which are computer-controlled stilts that allow virtual reality users to experience up and down steps [7].

2 THEORY

Existing medical studies have shown that patients who have limb-length discrepancy (LLD) demonstrate an altered gait pattern or a limp^[8]. Patients with Leg Length Discrepancy may usually be compensated by functional adaptation on the opposite side in an attempt to shorten the long leg [9]. Studies have shown that the gait of people with a limb difference greater than 2cm is very asymmetric, and the difference less than 2cm has no significant effect on the gait [10]. Based on this, we speculate that if people wear shoes with different heights (the difference of two shoes is less than 2 cm) and walk naturally in the virtual environment because the sensory receptors in tendons and joints produce hallucinations, they may cause people to orient incorrectly in the virtual environment, which can lead to an unconscious change in direction. To verify this hypothesis, we asked a shoemaker to

design an adjustable height shoe. The bottom height of this shoe can be adjusted manually to achieve the height difference between the two shoes. Its smallest adjustment unit is 0.2cm. We have designed two sizes of shoes: 240cm and 265cm, and we provided foam to solve the problem of oversized shoes. Then we designed an experiment to conduct a preliminary study of the role of foot height inconsistencies in the redirection.



Figure 1: Height-adjustable shoes in our experiment

3 METHOD AND MATERIALS

A total of 20 participants (12 males and 8 females) were recruited for the experiment. These participants were selected from a group of college students who did not have any self-reported Leg Length Discrepancy (LLD) or any abnormal pathology of lower limbs. The participants had an average age of 23.55 years old, an average height of 171.90cm, normal or corrected vision, and they all wore flat sports shoes to participate in the experiment.

The participants were asked to walk in a straight line, in a 6m by 9m room. And participants were told to stop as soon as they reached the destination. Participants wore a wireless HMD—Oculus Quest for the experiment. The difference in height between the two shoes is at least 0 cm and the maximum is 2.0 cm, and each adjustment unit is 0.2cm. So, this experiment designed and compared eleven conditions, each participant needs to repeat each height 5 times, a total of 55 trials. To reduce the impact of adapting to the environment, we conducted 4 sets of controlled experiments: 2(the shoe height changed from high to low X the shoe height changed from low to high) X2(left foot height changed X right foot height changed), 5 people in each group. By combining ascending and descending stimuli presentations, the detection threshold was better estimated.

Before experiment, participants were asked to put on an Oculus Quest and Adjustable Shoes. Then our staff will adjust the shoes to the corresponding height, and the participants didn't know about the change in shoe height. Participants experienced a 3D drawing game—Tilt Brush in a virtual environment. In this game, the virtual scene seen by the participants is blank. The participants were instructed to walk in a straight line in the virtual space and to hold the controller (the brush in the game) in front of their body as they walked. So, they can draw their tracks. It's important to note that the participants always looked forward as they walked, so they couldn't see the path drawn behind them. After each 5th walk, participants were asked to rate the comfort of the shoes.

Participants' actual time in the experiment was about 40-50 minutes. After the experiment, it took 2-5 minutes to complete the questionnaire and interview. And the simulator sickness was measured using the Kennedy-Lane Simulator Sickness

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Questionnaire (SSQ) which was administered immediately before the first and after the last trial [11].

4 RESULTS

Among the 20 participants, the first two participants had light leakage problems in their head-mounted displays. They didn't lean to either side under normal conditions, and they kept going straight throughout the experiment. After that, we used the nose pad for the HMD, and the situation was improved. Among the remaining 18 participants, they walked 5 times in the virtual environment with normal shoes. 9 of them tended to lean to the left, and 9 participants tended to the right. And the data of participants show that showed that the probability of the participant turning in the opposite direction as the shoe height increases in the experiment has nothing to do with the participant's leg length.

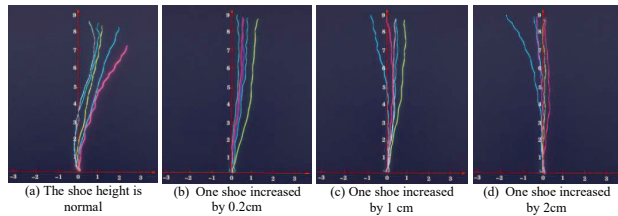


Figure 2: The Walking paths of P13. The plotted area covers 6m × 9m.

The biggest found of the paper was that as the height difference between the two shoes increased, the probability and angle of the participant's turning in the opposite direction also increased. For example, P13(Left foot height changed and shoes height changed from low to high) tended to walk in an arc to the right when he is wearing shoes of normal height in the virtual environment. As the height difference between the two shoes increases, he tended to walk in an arc to the left. Figure 3 shows the results of the experiment. The Probability's calculation formula is as follows:

$$P = N_{\text{(The number of walking in the opposite direction)}} / T_{\text{(Total times)}} \quad (1)$$

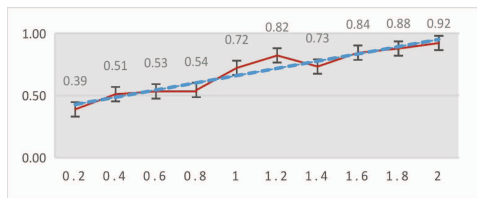


Figure 3: The relationship between the increased height of shoes and the change of walking direction. The x-axis represents the increased height of the shoe (Unit: cm), The y-axis represents the probability of walking in the opposite direction (compared with 5 walks with normal shoes)

In terms of user experience. The 5-level Likert scale was used to represent the comfort of the shoes, 0 is no feeling, 4 is very serious discomfort. When the height difference between two shoes was less than 0.6 cm, the participant had little discomfort with the height of shoes. When the height of a shoe increases, the discomfort perceived by the participants increases with the height. However, when the height difference between the two shoes decreased from high to low, the participants' discomfort with their feet showed an overall downward trend, but the fluctuation was small. Finally, comparison by means of a paired-sample two-tailed t-test revealed a significant difference between the overall simulator sickness scores obtained before and after exposure to

the virtual environment ($t(20) = -2.691$, $p < 0.05$). Average total SSQ score of participants before the experiment is about 65.2026, but the average total SSQ score of participants after the experiment is about 218.8274. It showed that the experiment can be carried out normally without notable simulator sickness.

5 CONCLUSION

The main contribution of this paper was the discovery of a new method -- an experiment using shoes with different heights. The most significant found in current research was that increasing the height of a single shoe can change people's walking direction in the virtual environment. When the shoe height was normal, the participants walked to one side. As the height difference between the two pairs of shoes the participant was wearing increased, the participant was more likely to lean on the other side. In the future, we can design an automatic heightening shoe that can meet the market demand, and through the combination of visual technology and adjustable height shoes, we can change the direction more flexibly in the virtual environment.

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